

# **A Survey on Underground Cable Fault Detection Systems (2022–2026)**

## **Abstract**

Underground power cables are increasingly used in modern power distribution systems due to their improved safety, reliability, and reduced environmental exposure. However, detecting and locating faults in underground cables is a complex task because the cables are buried beneath the ground and are not visually accessible. Traditional fault detection techniques are often time-consuming, costly, and inefficient. With the advancement of Internet of Things (IoT) technology, real-time monitoring and intelligent fault detection systems have emerged as effective solutions. This survey paper reviews and analyzes five research papers published between 2022 and 2026 that focus on IoT-based underground cable fault detection systems. The survey compares methodologies, sensor technologies, communication platforms, datasets, challenges, and future directions, providing a comprehensive understanding of the current research landscape.

## **1. Background**

Underground cables play a vital role in modern power transmission and distribution networks, especially in urban areas where overhead lines are impractical. Compared to overhead cables, underground cables offer advantages such as improved safety, reduced visual pollution, and protection from environmental hazards. Despite these benefits, underground cable systems face significant challenges when faults occur.

Faults in underground cables may arise due to aging, insulation failure, moisture ingress, construction activities, rodent damage, or mechanical stress. Locating these faults is difficult because the cables are hidden beneath the surface. Conventional fault detection techniques such as sectionalizing, thumping, and Murray loop methods require extensive manual effort and excavation, leading to increased downtime and repair costs. These limitations have motivated researchers to explore advanced fault detection techniques using IoT technology.

### 3. Overview of IoT-Based Cable Fault Detection

IoT-based underground cable fault detection systems typically consist of:

- **Sensors** (voltage, current, temperature, impedance)
- **Microcontrollers** (Arduino, PIC, NodeMCU)
- **Communication modules** (Wi-Fi, GSM, GPS)
- **Cloud platforms** (ThingSpeak, IoT dashboards)
- **Display/Alert systems** (LCD, SMS, email alerts)

These components work together to continuously monitor cable parameters, detect abnormal conditions, and notify maintenance personnel in real time.

## 4. Review of Selected Research Papers

### 4.1 Research Paper (2021 – Basic IoT Detection)

The 2021 study proposes an IoT-based underground cable fault detection system using voltage divider and resistance measurement principles. The system estimates the fault location by analyzing voltage variations across resistors placed along the cable. A microcontroller processes the data and displays the fault location on an LCD while transmitting information through IoT. This approach is simple and cost-effective; however, it is limited to basic fault types such as short circuits and lacks advanced analytics or predictive capabilities.

## **4.2 Research Paper (2022 – Multi-Sensor IoT System)**

The 2022 research introduces a multi-sensor IoT-based monitoring system that measures electrical parameters such as current and temperature. The collected data is transmitted to a cloud platform for real-time monitoring and analysis. This system improves detection accuracy and supports preventive maintenance by continuously observing system behavior. However, it is highly dependent on internet connectivity and does not address issues related to cybersecurity or data protection.

## **4.3 Research Paper (2023 – Wireless Sensor Network)**

The 2023 study focuses on integrating wireless sensor networks with IoT to enable faster fault detection and communication. The system monitors electrical parameters across multiple points and sends alerts to maintenance teams when abnormalities are detected. This approach significantly reduces fault response time and enhances system reliability. However, the deployment of wireless networks introduces complexity, and scalability remains a major challenge for large-scale implementations.

## **4.4 Research Paper (2024 – GSM + GPS System)**

The 2024 research presents an IoT-based underground cable fault detection system using Arduino, GSM, GPS, and cloud platforms such as ThingSpeak. The system accurately identifies the fault location and sends real-time alerts to maintenance personnel via SMS or cloud notifications. This improves practical usability and response efficiency. However, the system is primarily tested in simulated environments rather than real underground cable installations, limiting its real-world validation.

## **4.5 Research Paper (2025 – Predictive IoT System)**

The 2025 study introduces an advanced underground cable monitoring system that integrates IoT with predictive analytics and MATLAB simulations. The system enables proactive fault detection by analyzing trends and patterns in collected data. It supports real-time visualization and has the potential for future AI integration. Despite its advantages, the system is complex, requires high computational resources, and may involve higher implementation costs.

## **4.6 Research Paper (IoT-Based PAC Monitoring System – 2026)**

This research proposes an IoT-based monitoring system for precision air conditioning systems using sensors such as DS18B20 for temperature and PZEM-004T for current measurement . The system uses an ESP8266 microcontroller and transmits real-time data through a Telegram interface. It enables early fault detection by monitoring evaporator temperature and electrical current continuously. The system is low-cost and effective for preventive maintenance; however, it lacks predictive intelligence and is limited to monitoring rather than advanced fault analysis.

## **4.7 Research Paper (IEEE Survey on Fault Detection – 2026)**

This study presents a comprehensive review of underground cable fault detection techniques, including methods such as wavelet transform, time-domain reflectometry, and machine learning approaches . The paper analyzes various techniques based on accuracy, datasets, and performance metrics. It highlights the importance of intelligent fault detection systems and identifies challenges in existing methods. However, as a survey paper, it does not propose a specific implementation model.

## **4.8 Research Paper (ML + IoT EV Charging System–2026)**

The 2026 research introduces an IoT-enabled solar EV charging system integrated with machine learning for performance prediction . The system collects real-time voltage, current, and environmental data and uses machine learning models such as regression to predict power generation. This approach enhances system efficiency and enables intelligent decision-making. However, the system requires high-quality datasets and increases computational complexity.

## **4.9 Research Paper (Simulation-Based Fault Detection – 2024)**

This research focuses on simulation-based underground cable fault detection using tools such as MATLAB and Proteus. The system models different fault scenarios and analyzes system behavior under varying conditions. It provides a controlled environment for testing and evaluation. However, the results may not fully represent real-world conditions, limiting practical applicability.

## **4.10 Research Paper (Machine Learning-Based Fault Detection–2026)**

This study explores machine learning techniques for underground cable fault detection and classification. It uses historical and real-time datasets to train models for identifying fault patterns. The system improves accuracy and enables predictive fault detection. However, it requires large datasets, proper training, and computational resources, making implementation more complex compared to traditional methods.

## **5. Dataset Understanding**

The datasets used in the reviewed research papers are primarily sensor-generated datasets. These datasets include:

- Voltage levels
- Current flow
- Temperature variations
- Resistance and impedance values
- Fault distance and location
- Time-stamped real-time readings

The data is collected either through real-time IoT sensors or generated using simulation tools such as MATLAB. Cloud platforms store and visualize this data, enabling fault analysis, system evaluation, and decision-making. These datasets are essential for detecting faults, estimating fault locations, and improving system reliability.

## 6. Comparative Analysis

| No. | Year | Methodology                         | Sensors / Data Used           | Technology Used        | Key Features                               | Strengths             | Limitations                              |
|-----|------|-------------------------------------|-------------------------------|------------------------|--------------------------------------------|-----------------------|------------------------------------------|
| 1   | 2021 | Voltage divider & resistance method | Voltage                       | Basic IoT              | Fault detection using electrical variation | Simple, low cost      | Limited to basic faults, no intelligence |
| 2   | 2022 | Multi-sensor monitoring             | Current, Temperature          | IoT + Cloud            | Real-time monitoring                       | Improved accuracy     | Internet dependency, no prediction       |
| 3   | 2023 | Wireless Sensor Network             | Electrical parameters         | IoT + WSN              | Fast communication                         | Reduced response time | Scalability issues                       |
| 4   | 2024 | Arduino + GSM + GPS                 | Voltage, Current              | IoT + GSM + GPS        | Exact fault location, alerts               | Real-time alerts      | Mostly simulated system                  |
| 5   | 2025 | IoT + Predictive Analytics          | Multi-parameter               | IoT + MATLAB           | Predictive maintenance                     | High reliability      | High complexity, cost                    |
| 6   | 2026 | IoT PAC Monitoring                  | Temperature, Current          | IoT (ESP8266)          | Real-time monitoring via Telegram          | Low cost, practical   | No prediction capability                 |
| 7   | 2026 | Survey-based analysis               | Multiple datasets             | ML + Signal Processing | Comparative study of methods               | Wide analysis         | No implementation                        |
| 8   | 2026 | ML + IoT EV System                  | Voltage, Current, Environment | IoT + ML               | Power prediction                           | Intelligent system    | Requires training data                   |
| 9   | 2024 | Simulation-based system             | Simulated data                | MATLAB / Proteus       | Fault modeling                             | Safe testing          | Not real-world validated                 |
| 10  | 2026 | ML-based fault detection            | Historical + real-time data   | Machine Learning       | Fault classification & prediction          | High accuracy         | High complexity                          |

## **7. Challenges**

Despite significant advancements in IoT-based underground cable fault detection systems, several challenges still remain. These systems often require high initial installation costs due to multiple sensors and communication modules. Most solutions depend heavily on continuous internet connectivity, which limits their performance in remote areas. Additionally, many studies are tested only in simulated environments, reducing real-world reliability. Scalability is another concern, especially in large networks using wireless sensor systems. Cybersecurity risks are also largely unaddressed, making systems vulnerable to attacks. Furthermore, the lack of standardized datasets and the complexity of integrating advanced technologies such as machine learning create difficulties in implementation and performance evaluation.

## **8. Research Gaps**

The survey identifies several research gaps, including limited use of machine learning for predictive fault detection, insufficient real-time datasets from actual underground cable installations, lack of standard benchmarking techniques, and minimal focus on cybersecurity in IoT-based systems.

## **9. Correlation and Inferences**

The reviewed studies indicate a strong relationship between real-time sensor data and fault detection accuracy. Multi-sensor systems provide better fault diagnosis compared to single-parameter approaches, while cloud-based monitoring reduces fault response time. It can be inferred that integrating IoT with advanced analytics enhances system reliability and maintenance efficiency.

## **11. Conclusion**

This survey reviewed research papers on IoT-based underground cable fault detection systems and highlights the importance of real-time monitoring and improved accuracy. The findings show that IoT significantly enhances fault detection and reduces downtime. However, challenges such as scalability, data limitations, and lack of predictive intelligence still exist, indicating the need for more advanced and intelligent solutions in future systems.

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